A Comparative Framework for Diachronic Lexical Semantic Change in Brazilian Portuguese Political Discourse

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Abstract. *Shifts in political discourse often reflect deeper societal changes, but quantifying how specific terms evolve semantically over decades remains a significant challenge for computational linguistics. To compare existing approaches under different computational constraints, we propose a cost-aware, model- and language-agnostic comparative framework for lexical semantic change detection. We apply it to Brazilian Portuguese political discourse using a shared comparison panel of 55 lemmas combining drift candidates, stable controls, and theory-driven seeds. We compare three families of diachronic drift signals: TF-IDF lexical profiles, slice-specific Word2Vec models with Orthogonal Procrustes alignment, and a contextual BERT-based approach. The results show strong disagreement between the lighter baselines, with Spearman correlation of* -0.54 *and no overlap in their top-ranked terms, while contextual BERT shows weak positive associations with both cheaper methods. A filtered contextual panel reduces stable-control leakage and highlights terms such as "bloqueio", "salário", "intervenção", and "eleição". Rather than treating one detector as externally validated ground truth, this study presents an exploratory comparison of complementary drift sensitivities, interpretability, and computational cost in Brazilian Portuguese political discourse.*

1. Introduction

Political language carries historical change in its lexical choices, as shifts in ideological agendas, coalition structures, and policy framing leave traces in word usage over time. Analyzing these large-scale patterns by hand is impractical. Diachronic natural language processing (NLP) methods can capture some of the movement at scale, but different modeling choices foreground different aspects of change [Hamilton et al. 2016, Kutuzov et al. 2018, Tahmasebi et al. 2021]. In political corpora the picture is especially complex, since a term may drift because of a new policy referent, a rhetorical reframing, a shift in collocational profile, or a reorganization of discourse coalitions. No single mechanism accounts for all of them.

This complexity carries over to evaluation, where shared-task benchmarks have improved comparability in lexical semantic change detection [Schlechtweg et al. 2020] but curated external labels remain scarce, especially outside the few languages and corpora covered by standard tasks. Distributional pipelines can also react to temporal artifacts, alignment noise, and other confounds when outputs are read too literally [Dubossarsky et al. 2019]. Without externally validated ground truth, the central question becomes not which detector is "correct" but what each detector appears to capture when applied to the same political timeline.

This question is timely for Brazilian Portuguese. Large-scale political corpora such as BrPoliCorpus [Lima-Lopes 2025] are only now becoming available, yet comparative

drift studies that place multiple detection families side by side on the same Portuguese political material remain rare.

To address this gap, we propose a cost-aware comparative framework for diachronic semantic change analysis. The framework stages detection methods into lexical-statistical, static-embedding, and contextual tiers, then compares their outputs on a shared panel rather than letting each method define its own evidence base. We instantiate it on the floor subset of BrPoliCorpus, covering 24 yearly slices of Brazilian Chamber of Deputies speeches from 2000 to 2023. In this instantiation, TF-IDF, aligned Word2Vec, and BERTimbau Large are evaluated on the same 55 lemmas.

The paper addresses the following questions:

1. How strongly do the drift detection methods agree on the same lemmas?
2. Which lemmas identified as drift candidates are method-specific?
3. Does the higher-cost contextual stage of the proposed framework add enough interpretive value to justify its computational expense?

The rest of the paper follows that separation between design and instantiation. Section 2 positions the study in prior work, Section 3 defines the framework, Section 4 instantiates it on BrPoliCorpus, and Sections 5–6 analyze what the methods agree and disagree on.

2. Related Work

Distributional semantic change detection has evolved from co-occurrence comparisons to aligned static embeddings and then to contextual representations [Kutuzov et al. 2018, Tahmasebi et al. 2021]. Among static approaches, slice-specific Word2Vec models aligned with Orthogonal Procrustes have been shown to recover known historical shifts and reveal statistical regularities of change [Hamilton et al. 2016]. Such embeddings remain attractive because they are efficient and produce interpretable neighborhood movements, but they compress all usages of a word into a single vector, blurring cases in which older and newer usages coexist.

Evaluation remains a central challenge. Standardized shared tasks have improved comparability [Schlechtweg et al. 2020], yet many real-world studies still operate without external semantic ground truth, and apparent change can emerge from temporal referencing, alignment noise, or genre shifts rather than from semantic drift alone [Dubossarsky et al. 2019]. These concerns motivate the stable controls and agreement analysis we use below.

Contextual models provide a complementary perspective. BERT-style representations encode token-level usages and can surface variation that static spaces smooth over [Giulianelli et al. 2020, Devlin et al. 2019]. Scalable contextual methods that improve interpretability have also been proposed [Montariol et al. 2021], and pre-trained Portuguese checkpoints such as BERTimbau [Souza et al. 2020] have made contextual analysis feasible for Portuguese NLP.

Diachronic analysis of political language has gained traction through parliamentary corpora. The study of [Hofmann et al. 2020] compared lexical usage in Austrian political discourse across time, but comparable studies of Brazilian Portuguese political speech are

still scarce. Available Portuguese diachronic resources have focused primarily on historical and literary material rather than political discourse [Osório and Cardoso 2025]. BrPoliCorpus [Lima-Lopes 2025] now provides the scale needed for controlled comparative work in this domain.

This literature motivates a comparison in which no single detector is treated as ground truth. Our framework makes the comparison itself explicit: cheaper methods screen broadly, contextual modeling is used selectively, and agreement diagnostics show where the resulting rankings converge or diverge.

3. Comparative Drift Framework

We first describe the framework in general terms, independent of any particular corpus or model. Section 4 then instantiates it with BrPoliCorpus, TF-IDF, Word2Vec, and BERTimbau Large. Figure 1 summarizes the overall flow. Source code for the framework and experiments is available in the anonymous repository.

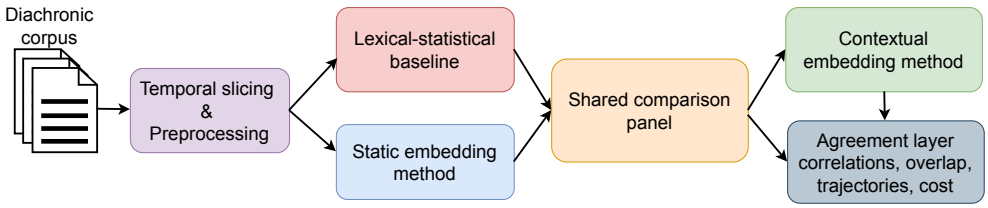

Figure 1. Staged comparative framework for lexical semantic change detection. Two cheaper tiers screen the scored vocabulary and nominate drift candidates; a shared comparison panel combines those candidates with stable controls and theory seeds before the expensive contextual tier and agreement analysis.

The framework assumes a diachronic corpus partitioned into temporal slices, where each slice must contain enough documents and tokens to support stable frequency estimates and distributional model training. A preprocessing step then normalizes surface forms to a shared lemma representation so that all three tiers operate on the same vocabulary. Not every word form is suitable for drift scoring, so the framework applies three joint filters before any method runs: a minimum per-slice frequency to guarantee sufficient token mass for embedding training, a document-dispersion threshold to prevent a single source from inflating apparent salience, and a minimum slice-presence rate to restrict attention to genuinely diachronic terms rather than episodic ones. Drift and stable-control candidates may be further restricted to dominant part-of-speech classes.

The framework then organizes lexical semantic change detection into three computational tiers arranged by increasing cost. A **lexical-statistical baseline** captures shifts in lexical salience with no model training. A **static-embedding method** aligns one vector space per temporal slice and measures neighborhood displacement. A **contextual-embedding method** represents sampled token usages with a pretrained language model, which can surface usage-level variation that cheaper type-level methods smooth over, but at much higher computational cost.

The key design principle is *staged cost awareness*. The two cheaper tiers first score the full eligible vocabulary, using their own representations to estimate diachronic drift

for each lemma. Their highest-ranked candidates feed into the expensive contextual stage, avoiding BERT inference over every eligible lemma. This ordering follows the practical distinction between broad type-based discovery and costlier token-based analysis in lexical semantic change work [Kurtyigit et al. 2021]. Cross-method comparison then requires a fixed evaluation set, so no method is judged only on the terms it would naturally select. The framework therefore prescribes a **shared comparison panel** assembled from four sources: (i) high-drift candidates nominated by the lexical-statistical baseline, (ii) high-drift candidates nominated by the static-embedding method, (iii) low-drift stable controls (lemmas ranked lowest by both cheaper methods), and (iv) theory-driven seed terms drawn from domain expertise or prior work. Because all three tiers score exactly the same candidate set, any disagreement reflects genuine differences in what each method measures rather than divergent vocabularies.

Finally, the framework adds an **agreement analysis layer**, which compares the rankings produced after all tiers score the same shared panel. This layer asks four practical questions. First, do two methods order the panel similarly overall? Pairwise rank correlations answer this at the level of the full ranking. Second, do the methods agree near the top, where candidate drift terms are selected? Top-k overlap curves track shared high-ranked lemmas as k grows. Third, do stable controls remain low ranked? Stable-control leakage rates identify controls that rise into candidate regions and therefore expose possible method-side noise, a risk emphasized by control-based analyses of semantic change models [Dubossarsky et al. 2017]. Fourth, after leaked controls are removed, which contextual candidates remain most useful for qualitative inspection? The filtered contextual ranking provides that final view.

The framework is model- and language-agnostic. Any combination of methods that respects the three-tier cost ordering can be substituted; the next section describes the particular choices used in this paper.

4. Methodology

We instantiate the framework on Brazilian Portuguese parliamentary discourse, specifying the corpus, preprocessing choices, and the model used in each tier.

4.1. Corpus and Preprocessing

A growing number of diachronic Portuguese resources are now available, as documented by [Osório and Cardoso 2025], ranging from syntactically annotated literary collections such as the Tycho Brahe Parsed Corpus [Galves and Faria 2017] to large-scale multi-genre repositories such as the Corpus do Português [Davies and Ferreira 2006]. However, these resources target historical-linguistic research across broad periods rather than domain-specific political analysis with continuous yearly slices. BrPoliCorpus [Lima-Lopes 2025] helps address this gap by providing large-scale parliamentary data for Brazilian Portuguese.

The experiments therefore use its subset containing Brazilian Chamber floor speeches from 2000 to 2023 organized into 24 yearly slices. Within BrPoliCorpus, the inaugural-speech subset reaches back to 1889 but contains only 35 documents, too few for stable distributional estimation. The floor subset offers the largest contiguous yearly coverage (428,366 speeches across 24 slices), a uniform parliamentary register, and exact date metadata, making it the most suitable segment for a year-by-year comparative drift study in the political domain.

As described in Section 3, not all tokens are eligible for drift scoring. Raw speeches are tokenized and tagged with `spaCy` (`pt_core_news_sm`). We retain nouns, verbs, and adjectives, exclude proper nouns, remove stopwords, punctuation, and numeric tokens, and keep only tokens with at least two characters. All text is lowercased with accents preserved. Each token is then mapped to its lemma form, producing the `content_lemma` representation used by all three tiers of the proposed framework (Figure 1). After preprocessing, the corpus retains 428,366 speeches and 63,036,642 tokens across 225,484 unique lemmas. Table 1 summarizes the corpus totals and the quartile structure of yearly variation.

A lemma enters the scored vocabulary only if it appears at least 50 times per slice, in at least 5 documents per slice, and in at least 80% of the 24 slices. These thresholds were fixed as practical corpus-stability defaults rather than tuned to optimize any particular method, and may vary when applied to corpora with different size, genre composition, or temporal density.

The frequency floor ensures sufficient token mass for reliable embedding training in each yearly slice; the document-dispersion requirement prevents a single speaker or parliamentary session from inflating a lemma's apparent salience; and the slice-presence constraint restricts attention to lemmas that are genuinely diachronic rather than episodic. Drift and stable-control candidates are further restricted to nouns and adjectives, and a small set of procedural high-frequency terms is excluded. These filters reduce the scored vocabulary to 3,221 lemmas for the two cheaper methods.

Yearly document counts range from 3,041 to 26,868 (median 16,752) and retained token volume from 630K to 4.2M (median 2.6M), which is why the frequency and slice-presence thresholds matter in practice.

Table 1. Corpus summary for the BrPoliCorpus floor subset. Yearly slice statistics are reported as the 25th percentile (Q1), median, and 75th percentile (Q3) across the 24 yearly files; token and character counts are shown in millions.

Period	Slices	Measure	Corpus total	Yearly slices		
				Q1	Median	Q3
2000–2023	24	Speeches	428,366	14,648	16,752	21,987
		Tokens	63.0M	2.00	2.62	3.45
		Characters	538.5M	16.9	22.3	29.7
		Lemmas	225,484	39,456	50,269	58,637

4.2. Drift Detection Tiers

Each tier is filled with a well-established model: TF-IDF for the lexical-statistical baseline, slice-aligned Word2Vec with Orthogonal Procrustes for the static-embedding tier, and BERTimbau Large for the contextual tier. These individual techniques are standard in the literature [Hamilton et al. 2016, Giulianelli et al. 2020]; the contribution here lies in combining them inside the staged comparison.

For the **lexical-statistical tier**, we compute a **TF-IDF** vector for each eligible lemma across yearly slices using smoothed inverse-document frequency. Drift is measured as the mean absolute change in TF-IDF weight between adjacent slices. This captures

shifts in lexical salience, reflecting how central a term becomes to each year's agenda, without relying on distributional neighborhoods.

For the **static-embedding tier**, we train independent **Word2Vec** (skip-gram) models per yearly slice (300 dimensions, window 5, 5 negative samples, minimum count 5, 5 epochs, 2 replicates with different seeds) and align them into a common space using Orthogonal Procrustes over the top 20,000 anchor words [Hamilton et al. 2016]. Drift is computed as the mean cosine distance between a lemma's aligned vectors in consecutive slices, averaged over replicates. This captures persistent neighborhood reorganization.

For the **contextual tier**, we embed sampled in-context occurrences using a Portuguese BERT checkpoint (**BERTimbau Large** [Souza et al. 2020]), extracting hidden states from layers -1 and -4. For each lemma and slice, up to 64 occurrences are sampled within a 48-token context window. The contextual drift score is computed as the mean cosine distance between slice-level centroid embeddings across consecutive slices. The agreement layer uses layer -1 as the primary ranking and layer -4 as a robustness check.

4.3. Shared Comparison Panel and Agreement Layer

Following the framework design (Section 3), we build one shared panel before running the contextual stage. The panel contains 55 lemmas: 15 Word2Vec-led drift candidates, 15 TF-IDF-led drift candidates, 20 stable controls, and 5 theory seeds. The method-led candidates are the highest-ranked nouns or adjectives from each cheaper tier after the eligibility filters described above. Stable controls are low-drift nouns or adjectives ranked near the bottom by both cheaper tiers, following the logic that control terms help reveal spurious drift signals rather than only highlighting positive examples [Dubossarsky et al. 2017]. The theory seeds are *democracia*, *corrupção*, *reforma*, *economia*, and *liberdade*, selected from the study design as broad political vocabulary anchors rather than as data-mined drift candidates.

The 15+15+20+5 composition is an empirical design choice, fixed before contextual scoring rather than tuned against results. We picked it by balancing three practical concerns: coverage of each method's top region, enough negative examples for the leakage diagnostics, and BERT inference time over 24 yearly slices. Fifteen candidates per cheaper tier stay within the informative top region of each method-specific ranking without inflating contextual cost, since each additional lemma requires sampled contexts across all slices. Twenty stable controls give the diagnostics enough negative examples to detect leakage, and five theory seeds act as an interpretive anchor without turning domain knowledge into a third candidate source. All three methods score this same panel. The agreement layer then reports pairwise Spearman rank correlations, top-k overlap curves for $k = 1, \ldots, 15$, stable-control leakage, and a filtered contextual ranking for qualitative inspection.

5. Results

Figure 2 reports the pairwise Spearman rank-correlation diagnostic from the agreement analysis layer. Each point is one of the 55 shared-panel lemmas, and each panel compares how two methods score the same lemma. A positive correlation means that two methods tend to rank the same lemmas as high-drift; a negative correlation means that a lemma ranked high by one method tends to be ranked low by the other. The four panels compare BERT vs. Word2Vec, BERT vs. TF-IDF, Word2Vec vs. TF-IDF, and BERT layers -4 vs.

-1. Layer -1 is the primary contextual ranking; layer -4 is used as a robustness view because upper BERT layers often carry semantic information while still differing in their sensitivity to surface form.

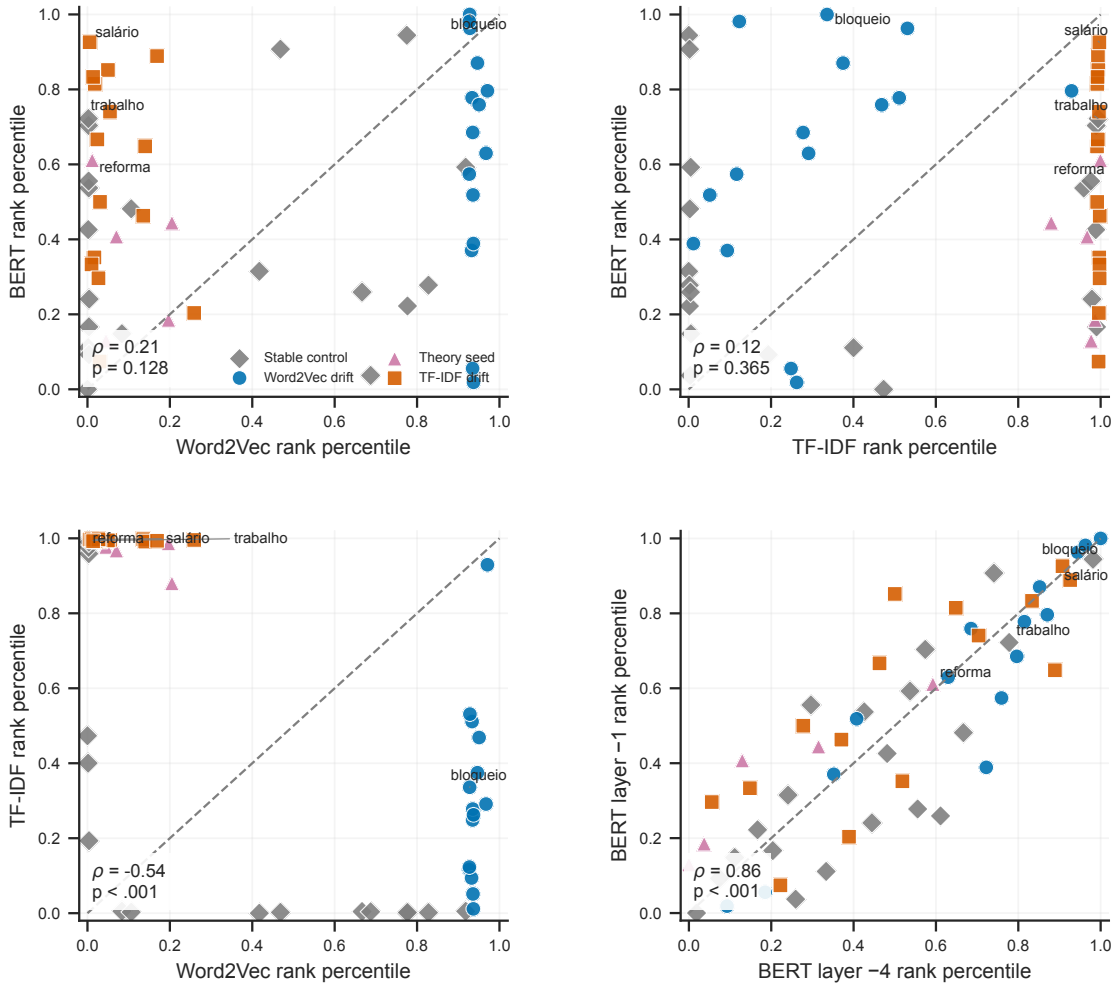

Figure 2. Pairwise rank agreement from the agreement analysis layer on the 55-lemma shared comparison panel across BERT vs. Word2Vec (top left), BERT vs. TF-IDF (top right), Word2Vec vs. TF-IDF (bottom left), and BERT layers -4 vs. -1 (bottom right). Points are lemmas from the shared panel; annotations report Spearman ρ and two-sided p-values.

The two lighter baselines disagree strongly even after they are constrained to the same candidate universe. As shown in Figure 2, the Word2Vec vs. TF-IDF panel displays a clear negative trend: lemmas ranked highly by one baseline tend to be ranked low by the other. Quantitatively, the two methods reach Spearman $\rho = -0.54$ ($p = 2.1 \times 10^{-5}$), confirming that they capture substantially different facets of change. Their top-15 overlap is zero, further showing that TF-IDF and Word2Vec favor markedly different drift candidates because they measure different types of change.

Figure 2 also shows that the contextual tier partly mediates this split without collapsing onto either baseline. In the BERT vs. Word2Vec and BERT vs. TF-IDF panels, contextual BERT displays weak but positive associations with both methods. On the

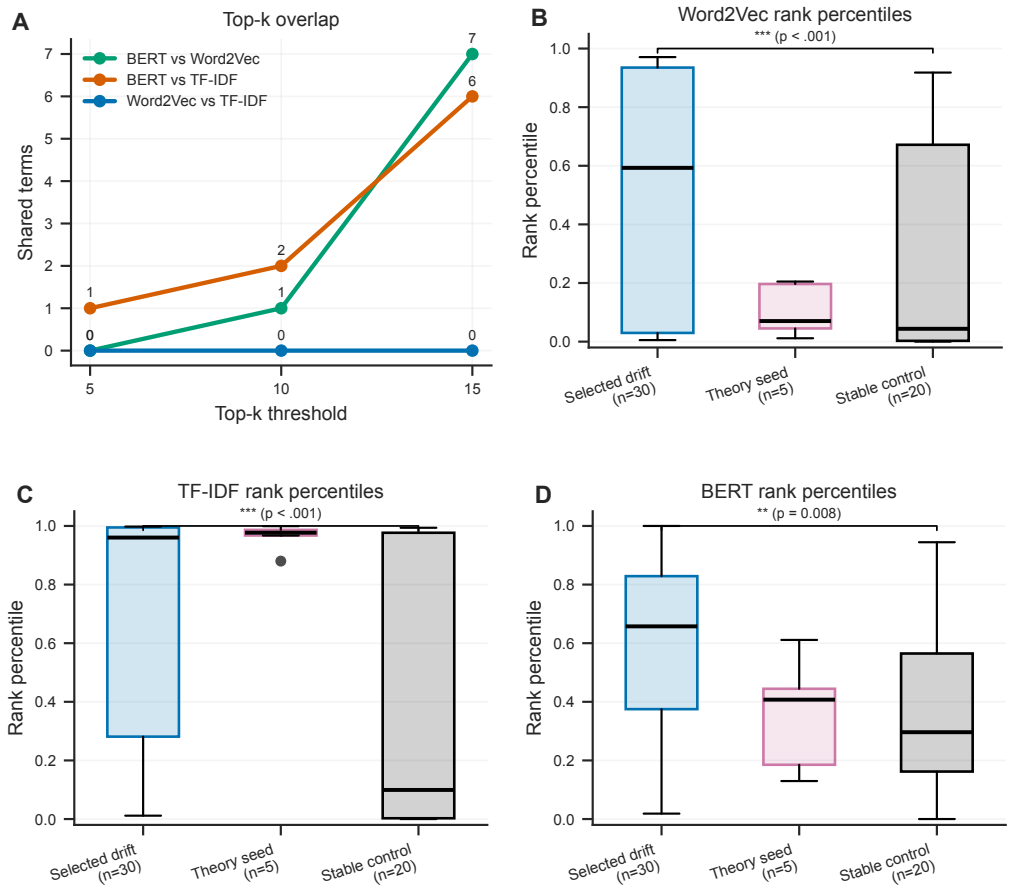

Figure 3. Overlap and rank-distribution summaries on the shared comparison panel. Panel A shows top-k overlap between methods. Panels B–D use standard boxplots to summarize rank-percentile distributions by bucket for Word2Vec, TF-IDF, and BERT respectively. Boxes show medians and interquartile ranges, whiskers extend to 1.5$\times$ the interquartile range, and points show outliers. Brackets report one-sided Mann–Whitney tests for drift terms vs. stable controls.

primary layer (-1), BERT reaches $\rho = 0.21$ ($p = 0.128$) with Word2Vec and $\rho = 0.12$ ($p = 0.365$) with TF-IDF. These associations point in the expected direction, but they are not statistically significant and are too small to justify treating contextual BERT as a substitute for either cheaper method. Instead, the contextual stage acts as an intermediate signal that partially aligns with both methods.

The BERT layer comparison gives a separate robustness check. Layers -4 and -1 agree closely (Spearman $\rho = 0.858$, $p < 10^{-15}$), so the contextual conclusions are not driven by a single extraction depth. This agreement is consistent with prior findings that mid-to-upper layers carry much of the semantic change signal, while final-layer representations can also encode surface-form information that biases change scores [Laicher et al. 2021, Periti and Tahmasebi 2024]. In this experiment, the similarity between layers -1 and -4 suggests that surface-form bias is not dominating the contextual ranking, although it cannot be ruled out.

Figure 3 reports the remaining diagnostics of the agreement layer. Panel A shows top-k overlap across methods, while Panels B–D use standard boxplots to compare the rank-percentile distributions of drift-candidate buckets and stable controls for Word2Vec, TF-IDF, and BERT.

Top-k overlap statistics reinforce the previous interpretation. In the primary BERT layer (-1), the top-15 overlap is 7 terms with Word2Vec and 6 terms with TF-IDF, whereas the baseline-to-baseline overlap remains zero. Thus, contextual BERT recovers candidates favored by both cheaper tiers while preserving an intermediate ranking structure of its own. The boxplots in Panels B–D further evaluate whether each method separates nominated drift terms from stable controls. Word2Vec and TF-IDF rank drift candidates significantly above stable controls ($p < .001$ in both cases, one-sided Mann–Whitney), showing clean bucket separation. BERT also separates the groups, but with weaker margins ($p = .008$). This weaker contrast is consistent with inspection of the raw contextual ranking, where some stable controls still appear near the top positions.

For that reason, the contextual ranking is most useful as a selective interpretive layer rather than as an unrestricted final candidate list. After filtering leaked controls, the remaining contextual candidates provide informative arbitration among terms proposed by the cheaper methods.

As a robustness check, we recomputed contextual scores using average pairwise distance (APD) [Giulianelli et al. 2020] instead of the centroid-based metric. APD operates on all individual occurrence embeddings rather than on slice-level centroids, giving equal influence to each usage regardless of embedding magnitude. On the primary layer, APD and centroid rankings correlate at Spearman $\rho = 0.79$ ($p < 10^{-12}$), with 7 of 15 top terms shared. The two metrics foreground different properties: APD promotes high-frequency polysemous terms whose embedding clouds are broad (e.g. *trabalho*, *público*), while the centroid metric favours terms with coherent directional shifts (*bloqueio*, *salário*). Notably, the centroid metric separates method-nominated drift candidates from stable controls more cleanly than APD ($p = .008$ vs. $p = .081$, Mann–Whitney). The strong overall correlation confirms that the comparative conclusions are robust to the choice of contextual aggregation, while the separation difference suggests that centroid distance is a less noisy signal for this panel-based comparison.

The agreement layer thus shows where the staged design pays off: the two cheaper tiers produce distinct candidate pools at low cost, and the contextual stage adds comparative evidence in the places where the baselines disagree.

6. Discussion

To examine how the methods differ at the level of individual terms, Figure 4 plots standardized temporal trajectories for four representative lemmas. Scores are normalized within each term and method so that the curves emphasize temporal shape rather than absolute magnitude, allowing direct comparison of when each method detects stronger movement. The selected cases cover four informative situations: a Word2Vec-led term with strong contextual support (*bloqueio*), a TF-IDF-led term with strong contextual support (*salário*), a theory seed whose contextual rank rises despite modest static drift (*reforma*), and a stable-control diagnostic term (*trabalho*) that reveals contextual leakage, where the model still ranks stable words as drift candidates.

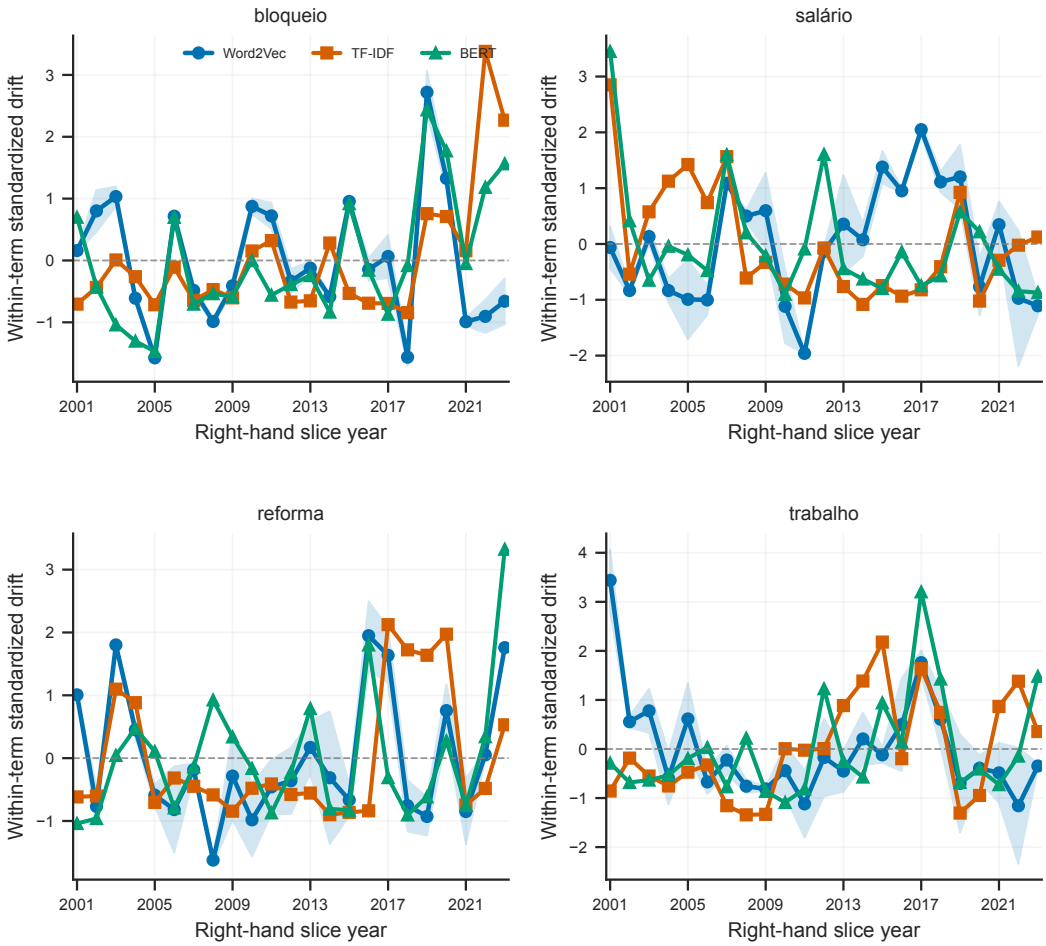

Figure 4. Representative term trajectories for *bloqueio* (top left), *salário* (top right), *reforma* (bottom left), and *trabalho* (bottom right). Scores are standardized within term and method to emphasize temporal shape rather than absolute scale; legend in the top-left panel applies throughout. Word2Vec ribbons denote replicate variability.

The disagreement itself is substantively informative, because each method foregrounds a different facet of lexical change. TF-IDF is most sensitive to lexical-profile reweightings, so a term whose salience rises in yearly agendas, policy disputes, or formulaic framing can register as drifting even if its local semantic neighborhood remains comparatively stable. Word2Vec, by contrast, captures persistent neighborhood reorganization across slices, highlighting terms whose distributional associations reorganize over time. Contextual BERT occupies an intermediate position and can support either a TF-IDF-led or a Word2Vec-led candidate through its token-level representations, although the stable-control leakage analysis shows that greater context sensitivity does not automatically produce a cleaner unrestricted ranking.

The term-level evidence supports this interpretation. *Salário* behaves like a lexical-profile case with strong contextual support, suggesting that both changes in agenda prominence and usage framing evolve over time. *Bloqueio* looks more like a neighborhood-displacement case in which the contextual tier aligns with the embedding-led signal.

Reforma illustrates a complementary pattern: it is only modestly ranked by the cheaper tiers, but it is ranked higher by the contextual model, suggesting that its change over time is better captured when context is taken into account. Finally, *trabalho*, included as a stable control, shows that the contextual model can still assign high drift scores to terms that are expected to be stable, which motivates the leakage diagnostics in the agreement layer.

These examples also clarify the value of the shared comparison panel. By including theory seeds and stable controls alongside method-led drift terms, the panel makes disagreements interpretable.

Table 2. Method scope and relative computational cost. The cost column reports order-of-magnitude class and approximate multiplier relative to TF-IDF, measured on the same single-machine run; absolute wall-clock times will vary with hardware.

Method	Signal focus	Scope	Relative cost
TF-IDF	Adjacent-slice salience shift	3,221 lemmas	seconds ($1\times$)
Word2Vec	Aligned neighborhood displacement	3,221 lemmas	minutes ($\sim 56\times$)
Contextual BERT	Usage divergence from sampled contexts	55 lemmas / 82,461 occurrences	hours ($\sim 600\times$)

Table 2 quantifies the cost gap. TF-IDF runs in seconds, Word2Vec in minutes at corpus scale, and contextual BERT is roughly two orders of magnitude more expensive even after restriction to the 55-lemma panel. Given the modest rank correlations reported in Section 5, the current evidence does not support replacing the cheaper methods with contextual models. Instead, it supports the staged design: cheaper methods screen the vocabulary broadly, while contextual modeling is applied selectively when disagreement is substantively interesting.

7. Conclusion

This paper introduced a shared, cost-aware comparative framework for contrasting three families of lexical semantic change detectors in Brazilian Portuguese political discourse. The research questions posed in the introduction receive three main answers from the evidence.

First, regarding the strength of inter-method agreement, TF-IDF and Word2Vec disagree strongly on the shared panel and contextual BERT remains only weakly correlated with either baseline, so the three families do not converge on a single ranking. Second, regarding method-specific candidates, the two cheaper methods identify largely non-overlapping sets of drift terms, indicating that each baseline highlights different aspects of lexical change. BERT partially bridges this split by placing candidates from both camps among its top-ranked terms, but it also retains some stable controls in high positions, so its raw ranking should not be treated as an unrestricted candidate list. Third, regarding the cost–benefit of the contextual stage, BERT provides useful adjudication when inter-method disagreement is substantively interesting, as illustrated by terms such as *bloqueio*, *salário*, and *reforma*. At roughly $600\times$ the cost of TF-IDF, though, the current evidence favors using it selectively rather than as a replacement for the cheaper methods.

These results frame disagreement itself as part of the contribution: the three method families respond to different mixtures of lexical salience, neighborhood displacement, and usage-level variation in Brazilian Portuguese political discourse. Because the comparative framework (Section 3) specifies roles and interfaces rather than particular models, it can be instantiated with alternative static or contextual embeddings, applied to other languages, or extended with additional tiers.

8. Limitations

The primary evaluative limitation is that this paper does not claim external semantic ground truth for BrPoliCorpus floor, so agreement and disagreement are analyzed as comparative evidence rather than as proof that one detector is correct. A second, methodological limitation is that the raw contextual ranking still admits some stable controls near the top, even though the filtered contextual list is useful for interpretation; contextual sensitivity does not automatically yield a cleaner unrestricted candidate list. The choice of contextual aggregation strategy also merits discussion. We report centroid-based drift as the primary contextual metric and verify it against APD [Giulianelli et al. 2020, Periti and Tahmasebi 2024] (Section 5); the strong correlation and the centroid metric's tighter bucket separation support this choice, though APD captures complementary embedding-cloud variance. Sense-based clustering methods [Montariol et al. 2021] were not attempted here. Finally, yearly slicing is appropriate for a first comparative baseline but compresses within-year shocks, campaign cycles, and short-lived institutional disputes that may matter for political language.

Future work could address the evaluative gap through external annotation or noisy supervision from labeled parliamentary datasets, though such resources still require careful deduplication before serving as reliable comparison targets. On the modeling side, adopting task-specialized models such as XL-LEXEME [Cassotti et al. 2023], exploring newer divergence metrics [Rachinskiy and Schlechtweg 2025], and applying orthographic-bias mitigation through subword masking [Laicher et al. 2021] would bring the contextual stage closer to current best practices. A symbolic companion analysis using lexical complexity or readability features could also help separate lexical-semantic drift from rhetorical or stylistic change.

9. Ethics Statement

The study analyzes public institutional political speech and reports only aggregate corpus statistics, ranked lemmas, and usage-level comparative signals. We do not profile individual speakers or infer private attributes. Even so, political corpora encode unequal representation and occasionally harmful rhetoric, so drift scores should not be read as normative judgments, evidence of speaker intent, or standalone claims about social meaning. All outputs are exploratory comparative signals whose interpretation depends on corpus composition, preprocessing choices, and the distinction between lexical salience, contextual variation, and broader discourse change.

10. Generative AI Use Disclosure

Generative AI tools were used to suggest alternatives for rewriting some sentences in a concise way and to correct spelling. All suggestions were manually verified by the authors.

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